

SMART REPLISHMENT MODEL 2.0

Our business problem was: Why do users churn, where does the customer journey break, and which product levers could improve retention and repeat purchasing?

Although recurring behaviour was not a significant churn differentiator, qualitative research revealed a concrete friction around repeat purchasing. We therefore explored Smart Replenishment as a friction-reduction opportunity rather than positioning it as a universal churn solution.

I synthesised the evidence after my analytics/validation phase.

And created:

Master Evidence → Key Quantitative Insights → Strategic Synthesis

And the key conclusion became:

There isn't one proven reason for churn.

Membership had a strong observed association with retention.

Recurring purchasing was real, but didn't meaningfully differentiate churn.

Coupons appeared more related to activity than churn.

Some hypotheses weren't supported.

It's my **strategic thinking phase**.

Now, product decision.

“Among the user problems surfaced in research, I chose recurring-purchase friction as a specific product opportunity that I could address.”

1. Product Problem

Routine Rebuyers already know what they want to purchase again, but they have to repeat the same search-and-add-to-cart process every time.

- Users repeatedly purchase the same products, such as their usual snacks/drinks or household essentials.

- The purchase itself isn't a discovery problem—the user already knows what they want.
- Repeating the same search and cart-building process creates unnecessary effort.

Why this is worth solving

If Zepto can recognize these recurring purchases and make the reorder process easier, it can reduce friction in repeat purchasing without forcing users to change how they shop.

2. Target User

Target User: The Routine Rebuyer

A Zepto user who repeatedly purchases the same products and shows a sufficiently predictable purchasing pattern.

Eligibility principle

Repeated + predictable → eligible for Smart Replenishment

Examples

Example 1 — Routine purchase

Coke + Chips purchased repeatedly → user shouldn't have to search for both every time.

Example 2 — Recurring essential

Milk / detergent / household essentials purchased repeatedly → user can quickly bring them back into the cart.

User Journey

Target User Scenario

Routine Rebuyer: A user who repeatedly purchases the same products and has a sufficiently predictable purchasing pattern.

Job to Be Done

When I need my usual products again, I want to reorder them without searching for and adding everything from scratch.

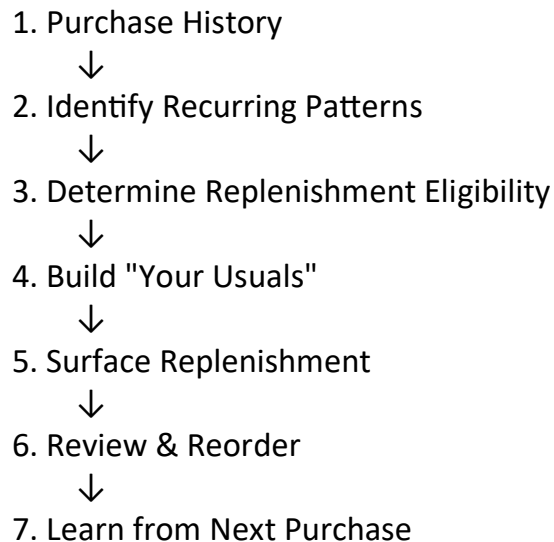
User Journey

Stage	User does	Zepto/system does	User value
1. Normal Purchase	Buys products as usual	Records purchase history	No change to existing experience
2. Pattern Detected	Continues purchasing the same products	Identifies repeated and predictable purchases	No manual setup required
3. Replenishment Surfaced	Returns to Zepto	Shows relevant “Your Usuals” products	Reduces effort of remembering/searching
4. Review Basket	Reviews, edits quantity or removes items	Pre-fills the suggested basket	User stays in control
5. Reorder	Confirms the basket	Places the order	Faster repeat purchase
6. Learn	Makes the next purchase	Updates the purchasing pattern	Future recommendations can improve

Core Loop

Purchase → Detect → Prepare → Review → Reorder → Learn

4. Product Flow — Smart Replenishment



CORE- Detect → Prepare → Reorder → Learn

Step 1 — Purchase History

The user continues using Zepto normally.

The system has access to the user's historical orders, including:

- what was purchased / category information available
- order dates
- order frequency
- order value
- other relevant order history

We are **not changing the normal shopping experience** at this stage.



Step 2 — Identify Recurring patterns

The system analyses the user's order history to identify recurring purchase patterns.

we'll investigate things such as:

- repeat purchases
- time between purchases
- recency
- consistency of the pattern

System asks: Is this recurring pattern predictable enough to be useful for a replenishment recommendation?

Repeated + sufficiently predictable + due/approaching expected purchase window

If yes → product becomes eligible for Smart Replenishment.

If no → nothing happens.

This is important because **we don't want to recommend every product the user has ever purchased.**

↓

Step 3 — Build “Your Usuals”

Eligible recurring purchases are grouped into the user's replenishment experience.

Example:

Your Usuals

Coke × 1

Chips × 1

Milk × 1

Important: For our current dataset, we can demonstrate this only at the **category level**.

The user doesn't have to manually search for these products.

↓

Step 4 — Surface the Replenishment Experience

When the user returns to Zepto, the product surfaces:

Ready to reorder your usuals?

Example-: Coke + Chips + Milk

Review & Reorder

We are deliberately keeping this **user-initiated**, rather than automatically ordering anything.



Step 5 — Review & Reorder

User can:

- change quantity
- remove an item
- add another item
- confirm the basket

Then:
Reorder



Step 6 — Learn from the Next Purchase

The new purchase becomes another data point.

Each subsequent purchase updates the user's observed purchase pattern and can improve future replenishment recommendations.

Each subsequent purchase becomes another data point that can improve future replenishment recommendation

For example:

Previous Coke interval: 8 days

New purchase: 10 days

The next recommendation can account for that new behaviour.

5. MVP Features — Smart Replenishment

Feature	What it does	Why it is needed
1. Recurring Purchase Detection	Analyses a user's order history to identify recurring purchase patterns.	Without this, we cannot identify Routine Rebuyers or their usual purchases.
2. Replenishment Eligibility Logic	Determines whether a recurring pattern is predictable/relevant enough to recommend.	Prevents irrelevant recommendations.
3. "Your Usuals"	Surfaces eligible recurring purchases in one place.	Removes the need to search for known purchases again.
4. Pre-filled Reorder Basket	Adds eligible usual purchases to a reviewable basket.	Removes repetitive add-to-cart work.
5. Edit & Review	Lets users change quantity, remove items or add products before ordering.	Keeps the user in control and avoids unwanted orders.
6. One-Tap Reorder	Allows the user to confirm the prepared basket with minimal effort.	Delivers the core value proposition: easier repeat purchasing.
7. Purchase Feedback Loop	Uses subsequent purchases as new behavioural data.	Allows future replenishment recommendations to improve.

So, **"What is the actual MVP?**

The MVP is a system that detects recurring purchasing patterns, prepares a personalized 'Your Usuals' basket, and lets the user review and reorder it with minimal effort. Subsequent purchases feed back into the system.

THE THINGS NOT ADDED:

Feature	Why not in MVP
Smart proactive reminders	We first need to validate whether users actually use replenishment recommendations.
Subscriptions / scheduled orders	Adds complexity around scheduling, skips, quantities and cancellations.
Automatic ordering	Too much loss of user control for the first version.
New membership product	Membership showed strong retention association, but we don't know the causal mechanism.
Advanced ML/AI prediction	Start with a simpler, explainable approach and validate the product before increasing model complexity.
Discounts for replenishment	Our analysis didn't establish coupons as a churn driver.

6. Product Logic & Edge Cases

6.1 Eligibility Logic

When should a user's purchase become eligible for Smart Replenishment?

The MVP would identify replenishment candidates using recurring purchase behaviour, purchase intervals and recency. A product/category becomes a candidate when the user's historical behaviour suggests that the purchase is repeated and reasonably predictable.

For the MVP, the system would evaluate four signals:

A. Repeat behaviour

Has the user purchased the same product/category multiple times?

B. Purchase interval

How much time usually passes between purchases?

Example:

8 days → 7 days → 9 days

This suggests a recurring pattern.

C. Consistency

Are the intervals reasonably stable?

Compare:

8, 8, 9, 7 days → predictable

vs.

2, 31, 5, 47 days → not predictable

D. Recency / due window

When did the user last purchase it?

If the typical interval is ~8 days and the user is approaching that window, the item/category becomes a candidate.

6.2 MVP Proposed Decision Logic

Repeated?

↓ No → **Don't recommend**

↓ Yes

Predictable interval?

↓ No → **Don't recommend yet**

↓ Yes

Approaching expected replenishment window?

↓ No → **Wait**

↓ Yes

Eligible for Smart Replenishment

Add to “Your Usuals” → User reviews → Reorders

The exact thresholds for repeat count, interval consistency and replenishment timing would need to be determined through historical-data testing and experimentation before production deployment.

6.3 What happens when behaviour changes?

Learning loop:

1. User buys earlier than expected

Example:

Expected: 8 days

Actual: 5 days

→ New purchase becomes a data point.

The next expected window can adjust.

2. User buys later

Expected: 8 days

Actual: 14 days

→ System updates the pattern rather than repeatedly recommending at day 8.

3. User skips

No purchase occurs.

→ Don't automatically assume they no longer want the product.

The system should wait for additional behaviour before changing its pattern significantly.

6.4 User Control

The MVP should never automatically place the order.

User can:

- remove an item
- change quantity
- add something else
- ignore the recommendation

6.5 Edge Cases

Situation	Product behaviour
Product bought only once	Don't recommend
Purchases are highly irregular	Don't recommend yet
User already bought the item recently	Don't surface immediately
User changes quantity	Record the new quantity/pattern
User removes an item	Don't force it back into the basket
User ignores recommendation	Wait for more evidence
User stops purchasing altogether	Reduce confidence / eventually stop recommending
Product/category unavailable	Don't recommend unavailable item; reassess when available

6.6 Important Data Limitation

category-level dataset → cannot prove SKU-level recommendation → production would require item-level data.

The current dataset is order/category-level and does not contain SKU-level purchase or quantity data. Therefore, this project proposes the replenishment logic but does not claim to have implemented or validated product-level recommendations.

7. Prioritization

If we cannot build everything, what do we build first?

Prioritisation is based on

Core user problem → **dependency** → **MVP feasibility** → **user control** → **validation need**.

P0 — Core MVP

Feature	Why
Recurring Purchase Detection	Identifies the user's recurring behaviour
Replenishment Eligibility Logic	Filters out irrelevant recommendations
Your Usuals	Surfaces the relevant purchases
Pre-filled Reorder Basket	Removes repetitive cart-building
Edit & Review	Keeps the user in control
One-Tap Reorder	Delivers the core convenience value
Purchase Feedback Loop	Allows future recommendations to incorporate new behaviour

P1 — Next iteration

Feature	Why
Smart Replenishment Reminders	Proactively brings the user back when they're likely to need their usuals
Replenishment Basket Improvements	Better grouping/personalization of recurring purchases

P2 — Future

Feature	Why
Subscriptions / Scheduled Replenishment	Useful for highly predictable consumption, but adds operational and UX complexity
Advanced Personalization	Improve recommendations once sufficient behavioural data exists
Membership Integration	Potential Premium-specific replenishment benefits, but the causal mechanism behind membership's retention association needs further investigation

8. Experiment & Metrics

If we launch Smart Replenishment, how do we determine whether it actually creates value? OR

Does Smart Replenishment increase repeat purchasing among eligible Routine Rebuyers?

Therefore, we build this around an **A/B test**.

Basic experiment

Control group:

Normal Zepto experience.

Treatment group:

Normal Zepto + Smart Replenishment.

Then compare their behaviour over a defined period.

Metrics

1. Primary Metric

Repeat Purchase Rate

Because the product's immediate job is to make repeat purchasing easier.

2. Secondary Metrics

- Replenishment recommendation → reorder conversion
- Time to reorder
- Orders/user
- Replenishment basket adoption

3. Guardrail Metrics

We don't want to improve orders by annoying people. So monitor:

- Recommendation dismissal rate
- Opt-out rate
- Unsubscribe/notification complaints if reminders are eventually introduced
- Cancellation/refund behaviour

So,

We need, Smart Replenishment

- easier repeat purchase
- more successful reorders
- potentially stronger repeat behaviour
- potentially better retention

8. Experiment & Metrics

8.1 Experiment Objective

The objective is to determine whether Smart Replenishment actually reduces friction in repeat purchasing and leads to more successful repeat orders among eligible Routine Rebuyers.

Product hypothesis:

If eligible Routine Rebuyers are given an easier way to reorder their usual purchases, they will be more likely to complete repeat purchases than similar users using the normal Zepto experience.

8.2 A/B Test Design

Group	Experience
Control Group	Normal Zepto shopping experience without Smart Replenishment
Treatment Group	Normal Zepto experience + Smart Replenishment

8.3 Experiment Flow

Control

Eligible Routine Rebuyer

- Normal Zepto experience
- User searches/adds products manually
- Order

Treatment

Eligible Routine Rebuyer

- Smart Replenishment surfaced
- “Your Usuals”
- Review/Edit
- Reorder
- Order

8.4 Primary Metric

Repeat Purchase Rate

The primary metric should measure whether users who receive Smart Replenishment are more likely to make another purchase during the defined observation window.

Immediate job of Smart Replenishment is:

Make repeat purchasing easier.

8.5 Secondary Metrics

Metric	What it tells us
Replenishment Recommendation → Reorder Conversion	Whether users actually act on the recommendation
Time to Reorder	Whether the product reduces the effort/time between expected replenishment and purchase
Orders per User	Whether repeat purchasing frequency changes
Replenishment Basket Adoption	Whether users engage with the prepared basket
Average Replenishment Basket Size	Whether users reorder multiple usual purchases together

8.6 Guardrail Metrics

Guardrail	Why monitor it
Recommendation dismissal rate	High dismissal may indicate irrelevant recommendations
Opt-out rate	Indicates whether users want to stop receiving the feature
Item removal rate	Shows whether the system is recommending unnecessary items
Cancellation / refund behaviour	Helps detect poor recommendations or unwanted purchases

8.7 Success Criteria

The experiment would be considered promising if:

1. **Repeat Purchase Rate increases** in the treatment group compared with control.
2. Users actually engage with the replenishment experience.
3. Replenishment recommendations convert into meaningful reorder behaviour.

4. Guardrail metrics do not show a significant increase in unwanted recommendations, cancellations or negative behaviour

8.8 What We Would Learn

There are three possible outcomes.

Outcome 1 — Positive

Treatment users show higher repeat purchasing with acceptable guardrails.

Decision: Continue the product and expand testing.

Outcome 2 — No meaningful improvement

Users see the feature but don't reorder more often.

Decision: Investigate whether the problem is recommendation quality, timing, basket composition or insufficient user value.

Outcome 3 — Negative

The feature creates irrelevant recommendations, high dismissal/removal rates or negative behaviour.

Decision: Do not scale it. Rework the eligibility logic or product experience.

8.9 Important Limitation

Because our current dataset is historical and observational, it **cannot tell us whether Smart Replenishment causes an increase in repeat purchases**.

The A/B test is therefore the proposed next step for causal validation.

Final experiment summary

Hypothesis: Making repeat purchases easier for eligible Routine Rebuyers will increase repeat purchasing.

Test: Randomized A/B test comparing normal Zepto experience with Smart Replenishment.

Primary KPI: Repeat Purchase Rate.

Secondary KPIs: Reorder conversion, time to reorder, orders/user and basket adoption.

Guardrails: Dismissal, removal, opt-out, cancellation/refund behaviour.

Decision: Scale, iterate or stop based on incremental impact and user-quality signals.

9. Roadmap

Phase 1 — MVP: Reduce Reorder Friction

Goal: Make repeat purchases easier for Routine Rebuyers.

Build:

- Recurring purchase pattern detection
- Replenishment eligibility logic
- **Your Usuals**
- Pre-filled reorder basket
- Review/edit
- One-tap reorder
- Purchase feedback loop

Success:

Validate whether easier reordering increases repeat purchasing without creating irrelevant recommendations.

Phase 2 — Proactive Replenishment

Goal: Move from users discovering their usuals to Zepto proactively helping them at the right time.

Build:

- Smart replenishment reminders
- Personalized replenishment timing
- Better recommendation ranking
- User reminder preferences
- More personalized replenishment baskets

Phase 3 — Recurring / Scheduled Purchases

Goal: Support highly predictable consumption patterns.

Build:

- Scheduled replenishment
- Flexible subscriptions
- Skip / pause
- Quantity adjustments
- Rescheduling
- Cancellation controls

Potential use cases:

- diapers
- baby products
- household essentials
- other highly predictable recurring needs

Phase 4 — Personalization & Membership Integration

Goal: Make replenishment increasingly personalized and potentially connect it with Zepto's broader retention ecosystem.

Potential features:

- More advanced demand prediction
- Personalized replenishment recommendations
- Better handling of changing purchase patterns
- Premium-specific replenishment benefits

Roadmap Summary

Phase	Focus	Core outcome
Phase 1	Smart Replenishment MVP	Reduce repeat-purchase friction
Phase 2	Proactive Replenishment	Remind users at the right time
Phase 3	Scheduled Purchases	Automate predictable recurring needs

Phase	Focus	Core outcome
Phase 4	Personalization + Membership	Build a broader retention ecosystem

I deliberately started with assisted replenishment rather than subscriptions or fully automated ordering. I wanted to validate the core behaviour—whether reducing reorder friction actually increases repeat purchasing—before adding more automation and personalization.